

FACULTY OF COMPUTING AND TECHNOLOGY UNIVERSITY OF KELANIYA

A SMART EMISSION MONITORING SYSTEM FOR FACTORIES PROJECT PROPOSAL

- **GROUP NUMBER:** CT-19
- **COURSE:** GTEC 23032- PROJECTS IN TECHNOLOGY II
- **SUPERVISOR:** DR. W.D. CHANAKA UDAYANGA
- **GROUP MEMBERS:** CT/2021/034 - KAFEELA U.L
CT/2021/080 – ABISEKAN. A
CT/2021/081- NIRUTHTHIGAN. M
CT/2021/086 – KIROJAN. S
CT/2021/088 – TISHANY. T

Table of Contents

Page No

Introduction.....	03
Literature Survey.....	04
Objectives	05
Methodology	06
Timeline	07
Budget.....	08
References.....	09

Introduction

Our project aims to develop a smart web application that can detect and track harmful gas emissions from factories in real-time. If a factory exceeds the legally allowed emission levels, the system will automatically send a notification via email or SMS to the factory owner and government.

This ICT-integrated solution will provide an automated, efficient, and transparent way to track emissions. It will help industries follow environmental regulations, reduce pollution, and support global efforts to combat climate change. By using technology to improve environmental monitoring, this project will contribute to a cleaner, healthier, and more sustainable future.

Literature survey

Many researchers have explored gas emission monitoring systems to help industries reduce pollution. Several studies discuss IoT-based solutions that use gas sensors to detect harmful emissions in real-time. These systems allow industries to track emissions, store data, and receive alerts when pollution levels exceed safety limits. However, some existing solutions are expensive or lack real-time alerts, making them less practical for large-scale use.

Another key area of research is Energy Management Systems (EMS), which help industries optimize power consumption and lower carbon emissions. A study on low-carbon concrete production showed how big data analytics can help industries reduce pollution. Wireless Sensor Networks (WSNs) have also been tested for air pollution monitoring, proving to be cost-effective and highly accurate. Recent research focuses on AI and machine learning to improve predictive monitoring. Tools like Carbon Tracker help industries measure and reduce their carbon footprint. Another study on industrial boilers in China found that advanced filtration and pollution control technologies can effectively lower emissions.

Overall, these studies show that IoT, AI, and big data analytics are essential for modern gas emission monitoring. Future research should focus on enhancing sensor accuracy, integrating AI for predictive monitoring, and improving real-time alert systems to help industries follow regulations and protect the environment.

Objectives

1. Develop a smart web application that can detect and track harmful gas emissions from factories in real-time.
2. Use gas sensors to measure emissions and store data in a centralized database.
3. Automatically send notifications via email or SMS to factory owners and government officials if emission levels exceed legal limits.

Methodology

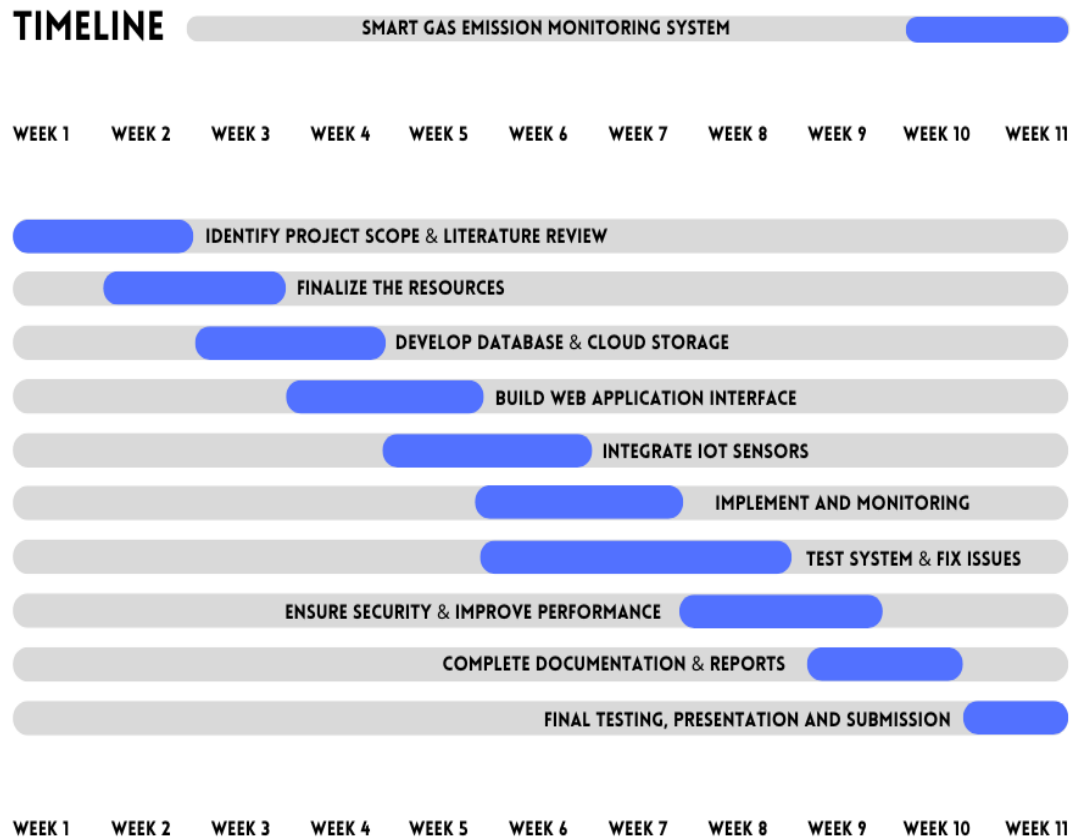
We are developing a web application to monitor harmful gas emissions from factories. First, we gather information from factory owners, managers, and government officials to understand the requirements and regulations for emissions. Once we have a clear understanding, we move on to designing the system.

The user interface will be built using HTML5, Bootstrap5, and React.js to make it easy to use and responsive. The backend will be developed using Node.js and Express.js to handle data processing and server-side operations. MySQL will be used to store emissions data and user information in an organized manner.

After completing the design, we begin implementing the system. This includes coding the frontend and backend, connecting gas sensors to collect real-time emission data, and setting up Firebase Authentication to secure user access. The system will automatically send notifications via email or SMS if emissions exceed legal limits.

Once the system is built, it will be tested to ensure everything functions properly. After testing, the application will be deployed for use in factories, and training will be provided to users. Finally, the system will be maintained and updated regularly to ensure it continues to meet regulatory requirements and provide accurate data. Therefore, we are planning to do our project through Waterfall methodology.

Timeline



Budget

Item	Quantity	Unit price	Price
Co ₂ sensor (MG811)			
Co Sensor (MQ-7)	1	430.00	430.00
ESP32 (microcontroller)	1	1650.00	1650.00
Bread board	1	580.00	580.00
LED	3	15.00	45.00
Jumber wire	1 bundle	240.00	240.00
Total			2,945.00

Budget Considerations

For our prototype, we will monitor CO (Carbon Monoxide) levels because CO₂ (Carbon Dioxide) sensors are too expensive (11900.00) for this stage. However, in a real factory application, we would use CO₂ sensors to monitor both gases. We plan to add CO₂ sensors when the project is fully developed, and funding is available.

References

1. Martillano, Dennis A., et al. "Android Based Real-Time Industrial Emission Monitoring System Using IoT Technology." *J. Commun.* 12.11 (2017): 623.
2. Liu, Weihua, et al. "The potential of smart factories in reducing environmental emissions: the evidence from Chinese listed manufacturing firms." *Humanities and Social Sciences Communications* 11.1 (2024): 1-16.
3. Kim, B., & Jeong, J. (2022). Real-time low-carbon prediction in ready-mixed concrete production process for smart manufacturing. *Procedia Computer Science*, 203, 205-212.
4. Tan, Y. S., Ng, Y. T., & Low, J. S. C. (2017). Internet-of-things enabled real-time monitoring of energy efficiency on manufacturing shop floors. *Procedia CIRP*, 61, 376-381.
5. Simbeye, D. S. (2017). Industrial air pollution monitoring system based on wireless sensor networks. *Journal of Information Sciences and Computing Technologies*, 6(2), 612-624.
6. Artanti, Lutfia Nisa, et al. "Smart industrial fan monitoring system for the implementation of sustainable development goals." *Proceedings ASTRAtch International Conference*. Vol. 1. No. October. 2024.
7. Syafrudin, M., Alfian, G., Fitriyani, N. L., & Rhee, J. (2018). Performance analysis of IoT-based sensor, big data processing, and machine learning model for real-time monitoring system in automotive manufacturing. *Sensors*, 18(9), 2946.
8. Lamprinos, Ilias, Marios Charalambides, and Michael Chouchoulis. "Greenhouse monitoring system based on a wireless sensor network." *Proceedings of the 2nd International Electronic Conference on Sensors and Applications*. 2015.
9. Bachram, Heidi. "Climate fraud and carbon colonialism: the new trade in greenhouse gases." *Capitalism nature socialism* 15, no. 4 (2004): 5-20.

10. Patil, Mrs Ashwini S., and S. R. Kumbhar. "A Review on Industrial Emission Monitoring System." *International Research Journal of Engineering and Technology (IRJET)* 7.07 (2020): 2395-0056.